IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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Executive Summary

- The SpaceX launch data was obtained on SpaceX and Wikipedia through webscraping
- The data was then updated to a dataframe, filtered to include Falcon 9 launches, and cleaned such that missing values for Payload Mass were replaced with the mean
- Data exploration was performed through SQL to obtain information such as successful and failed landing attempts
- An interactive map was created using Folium to visualize launch sites, their launches, and their surroundings such as coastlines and highways
- Plotly was used to generate a dashboard featuring a range slider which demonstrates failures and successes between booster versions; additionally, the dashboard featured a pie chart mapping success and failure rate
- Finally, four models: logistic regression, classification trees, supervised model, and K-nearest neighbors. Each model was tuned with GridSearch to find the best hyperparameters.
- The results showed that landing results with the SpaceX Falcon 9 can be predicted best with the classification tree model. Though logistic regression and supervised models performed equally as well as the classification tree model on test sets with 83.33% accuracy, the classification tree model performed better than the rest on the training set with an accuracy of 87.5%.

Introduction

- SpaceX's Falcon 9 is reusable, which cuts the cost of rocket launches
- Determining probability of the rocket launching or not can assist in predicting costs
- The goal of this project is to determine if, and how accurately, the landing outcome of SpaceX Falcon 9 launches can be predicted

Section 1

Methodology



Methodology

Executive Summary

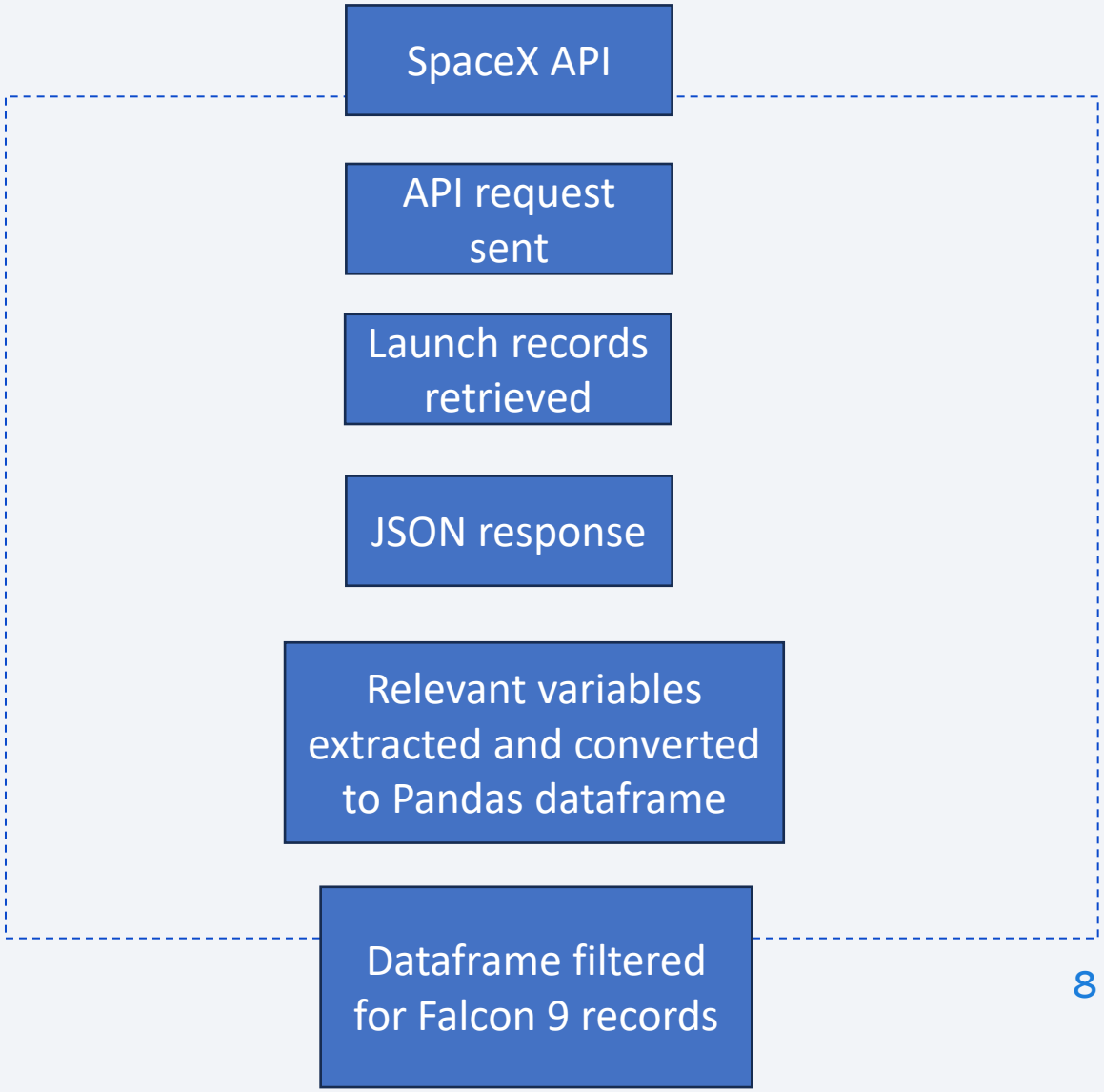
- Data collection methodology:
  - Data was collected through webscraping
- Perform data wrangling
  - Data was formatted into a Pandas dataframe and filtered to only include Falcon 9 launches
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Predictive analysis was developed using four different models: classification trees, supervised learning, logistic regression, and K-nearest neighbors, to determine which produced the most accurate results on test data after being trained on other data

Data Collection

- Data was collected by calling to the SpaceX data API, parsing, and then appending data to lists
- The response was made to a Json file and then converted into a Pandas dataframe
- The API was called to store data into the lists
- A new dataframe was created which filtered out only Falcon 9 results
- Missing values for payload mass were replaced with the mean

Data Collection – SpaceX API

- Capstone_Project/jupyter-labs-spacex-data-collection-api.ipynb at main · alyd5827/Capstone_Project

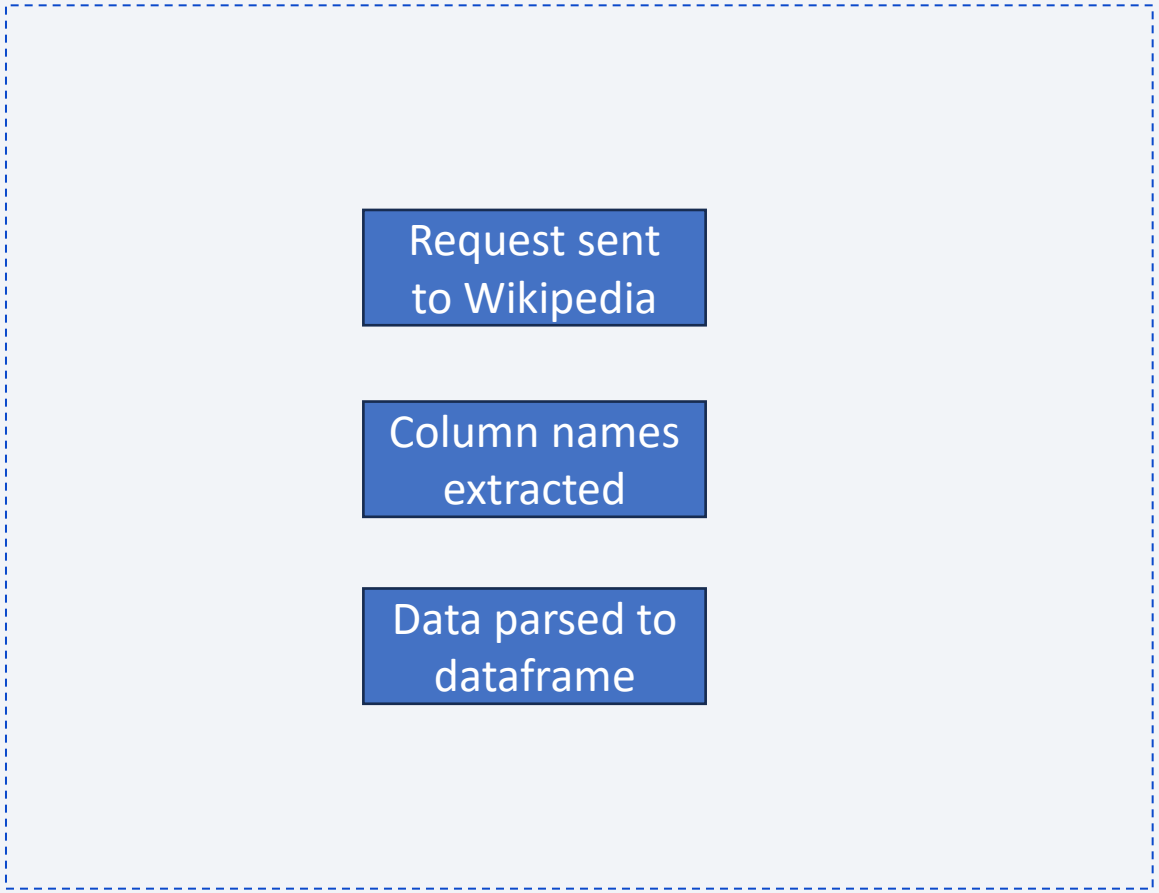


Data Wrangling

- A GET request was sent to the Falcon 9 Wikipedia launch page
- All column names were extracted from the HTML table header
- Data was parsed from the tables and converted to a Pandas dataframe

Data Collection - Scraping

- [Capstone_Project/jupyter-labs-webscraping.ipynb at main · alyd5827/Capstone_Project]



EDA with Data Visualization

- Scatterplot charts were generated to visualize the relationships between flight number and launch site, payload mass and launch site, flight number and orbit type, and payload mass and orbit type
- A bar chart was created to explore the relationship between success rate and orbit type
- A line chart displayed the relationship between success rate and year to see the progression on successful landings

- Capstone_Project/edadataviz.ipynb at main · alyd5827/Capstone_Project

EDA with SQL

- Viewed launch sites and filtered information by launch site
- Calculated total payload mass carried by boosters
- Calculated average payload mass carried by F9 v1.1 booster
- Viewed and counted successful and failed launches

- Capstone_Project/jupyter-labs-eda-sql-coursera_sqllite.ipynb at main · alyd5827/Capstone_Project

Build an Interactive Map with Folium

- Markers and circles were added to mark launch sites
- Lines were used to identify distance from major areas such as coastlines and highways
- Objects were added to visualize launch site proximity to highways, railways, coastlines, and major cities

- Capstone_Project/lab_jupyter_launch_site_location.ipynb at main · alyd5827/Capstone_Project

Build a Dashboard with Plotly Dash

- A pie chart was made to display failure and success launch outcomes for the CCAFS SLC-40 launch pad to determine the success rate of launches
- A range slides was created to display payload mass and success by booster version to see which payload masses and booster versions were successfully launched

- Capstone_Project/spacex-dash-app.py at main · alyd5827/Capstone_Project

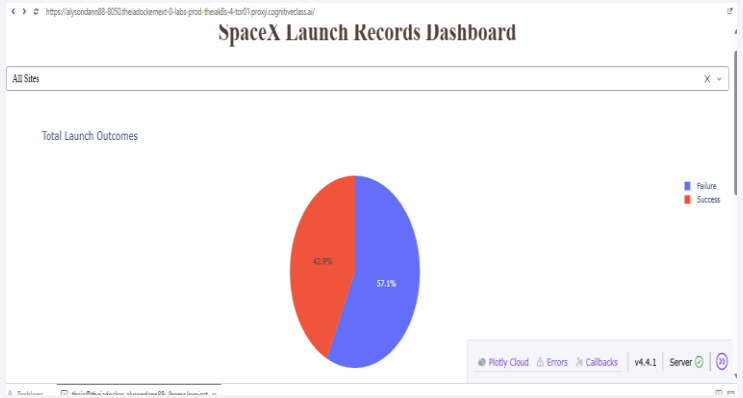
Predictive Analysis (Classification)

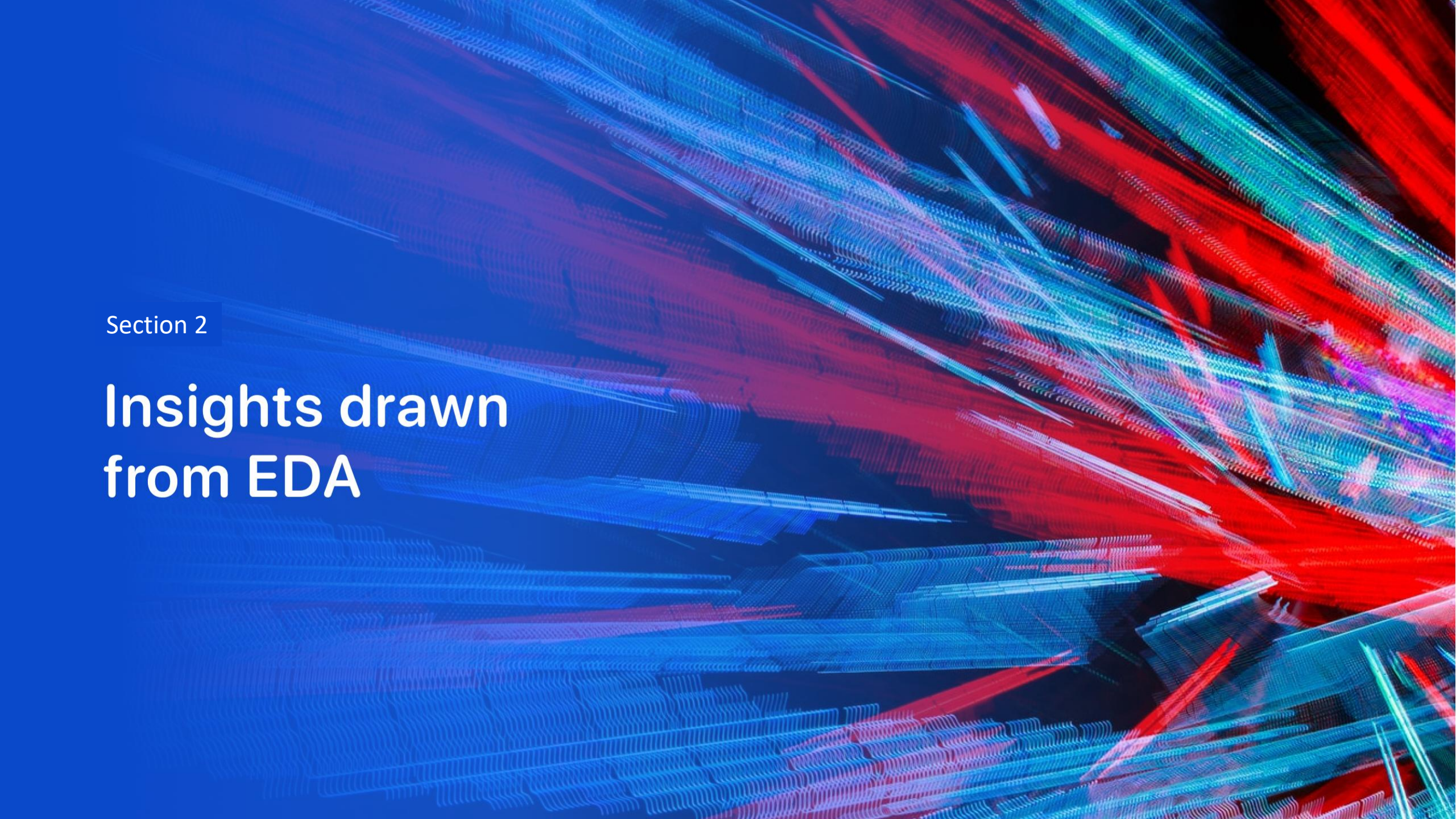
- The data was standardized and split into training and testing sets, with 20% allocated to testing
- Four classification models (logistic regression, supervised model, tree classifier, and k-nearest neighbors) were used on the data
- A grid search was performed on each classification model to determine the best parameters, and accuracy was evaluated based on the given best parameters
- Confusion matrices were populated for each model to visualize accuracy
- Test data was used on all models, and accuracy was found for each

- Capstone_Project/SpaceX_Machine Learning Prediction_Part_5.ipynb at main · alyd5827/Capstone_Project

Results

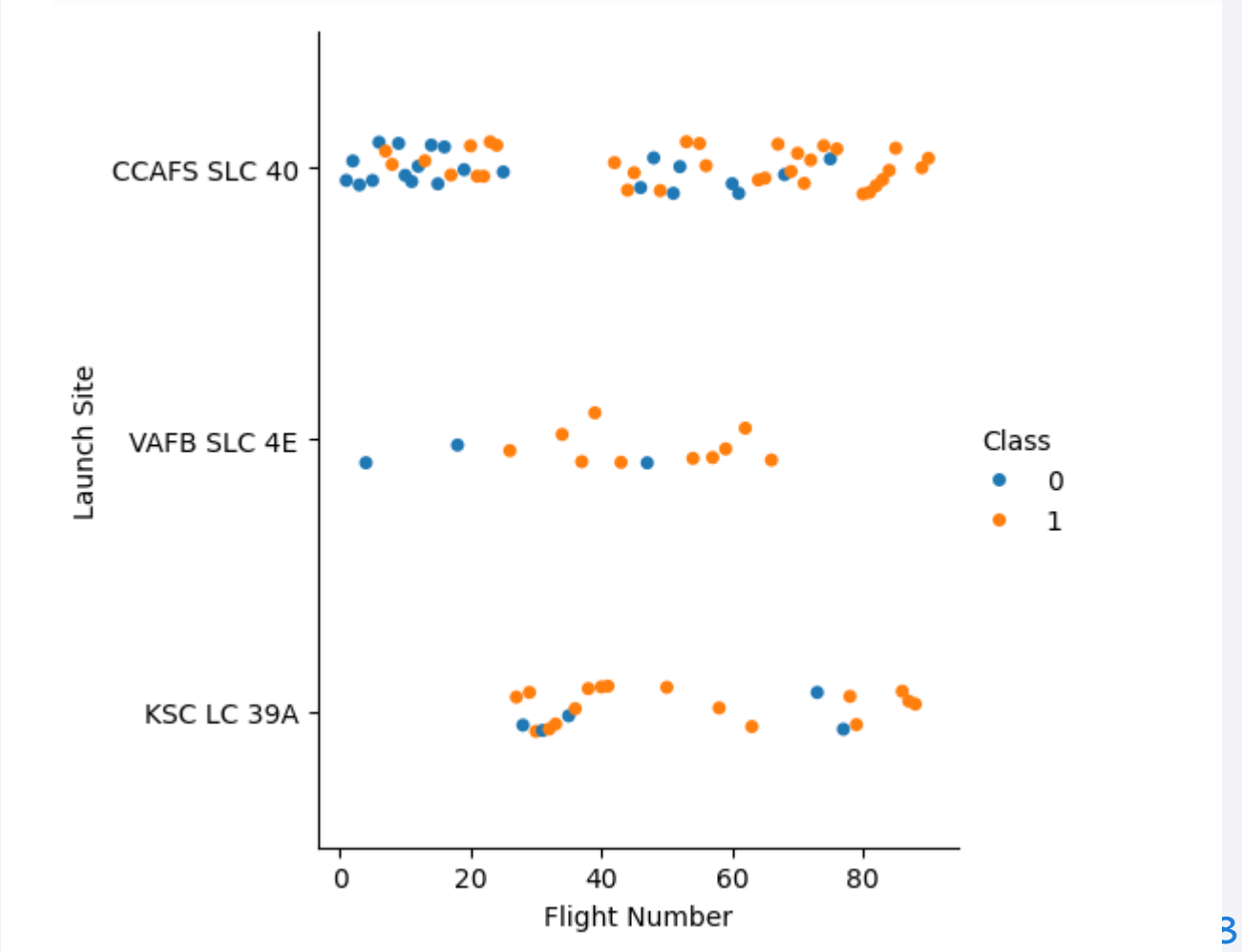
- Exploratory data analysis results:
  - Exploratory data analysis found a sharp increase in successful landings after the year 2013 with a slight decrease in 2018
  - Exploratory scatter plots did not show any obvious relationships, though higher payload mass may improve success rates based on the payload
  - The bar chart comparing orbit and success rates gave insight into orbits with the highest success rates, some of which were 100%
- Predictive analysis results:
  - Predictive analysis performed on the test sets showed 83% accuracy on all models except K-nearest neighbors, which instead had an accuracy result of 72%
  - The best model is supervised model which has an accuracy of 83.33% on test data and 84.82% on training data




Section 2
Insights drawn from EDA

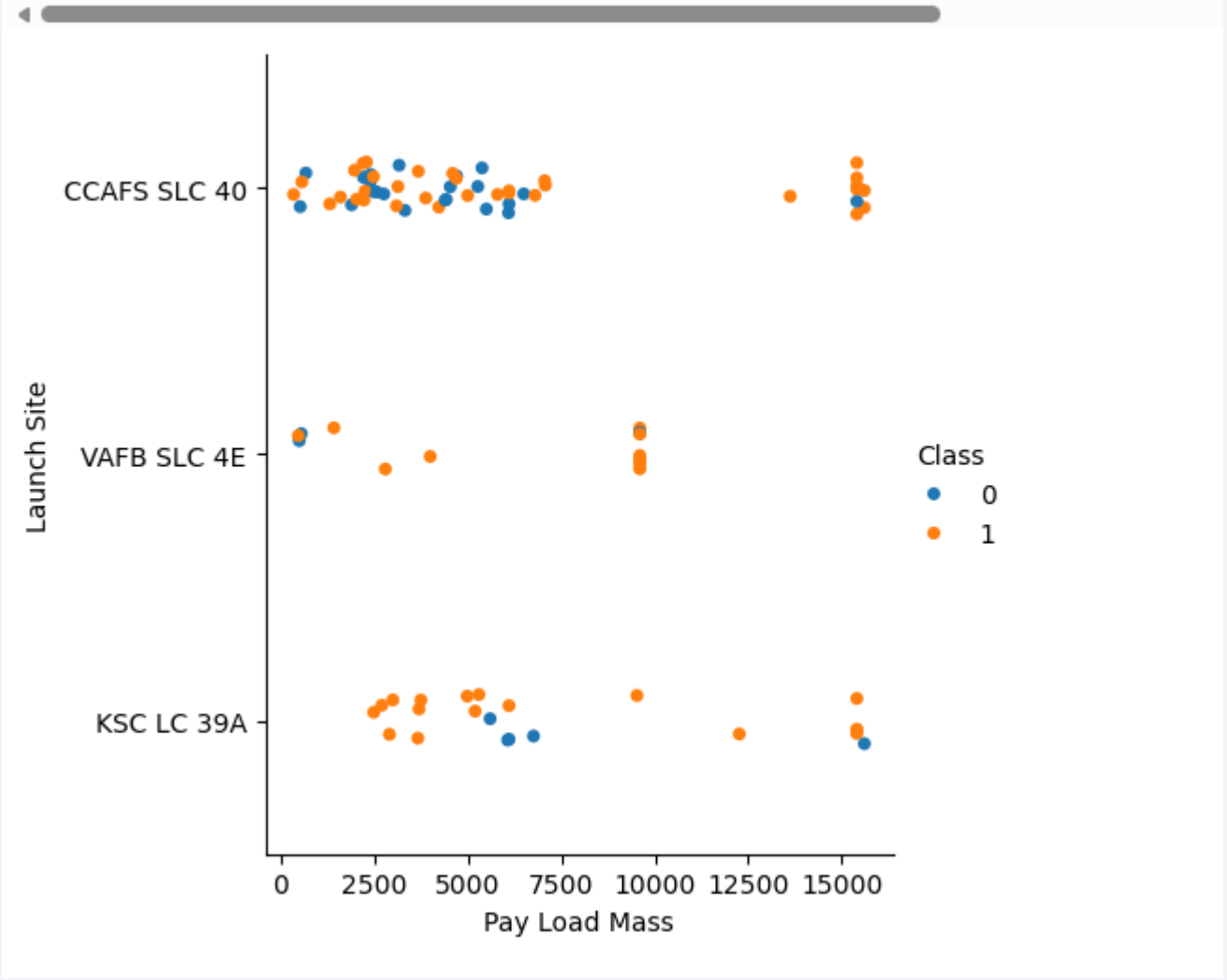
Flight Number vs. Launch Site

- The scatter plot displays success and failures by flight number and launch site
- Orange is class 1 which indicates a successful launch and blue is class 0 which indicates a failed launch
- There appears to be successful launches at higher flight numbers
- The CCAFS SLC 40 seems to have the worst success rate out of the launch sites



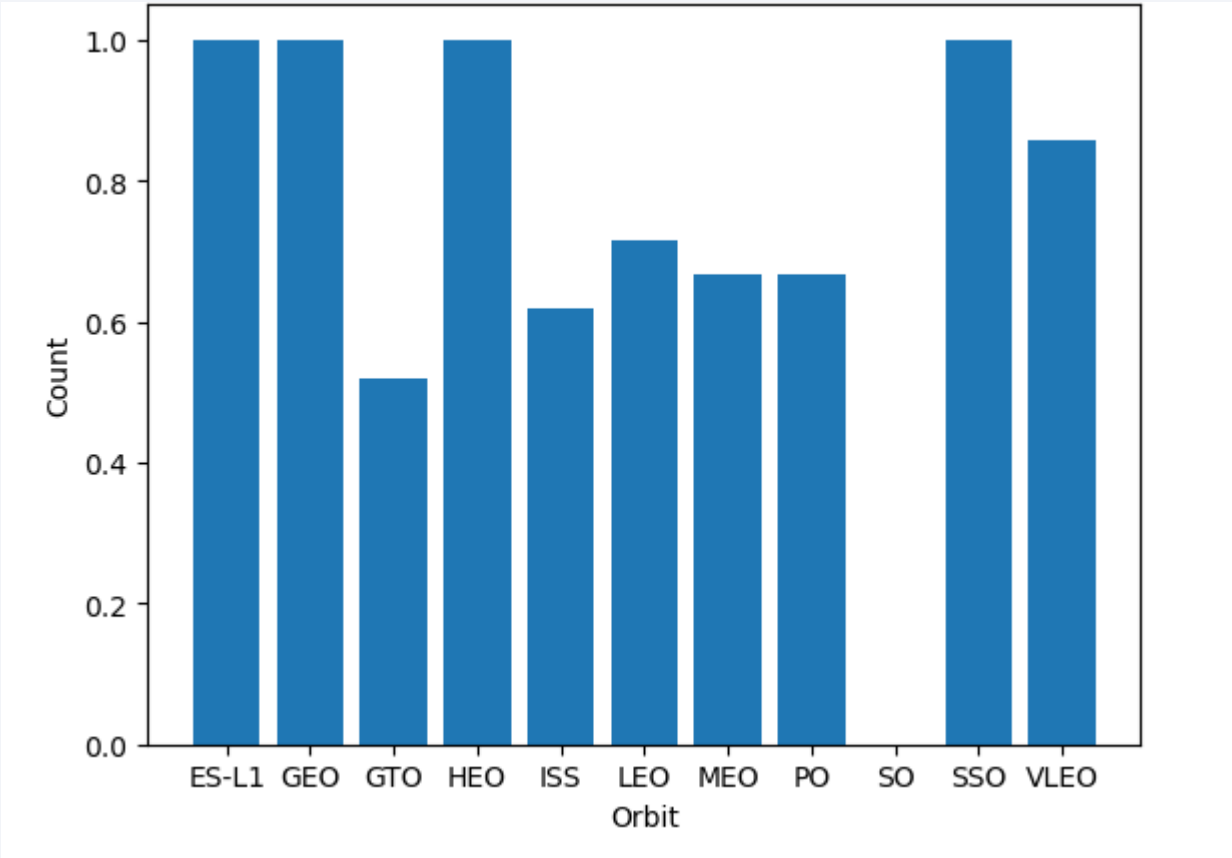
Payload vs. Launch Site

- The scatterplot depicts success and failures by payload mass and launch site
- A higher payload mass appears to increase the chances of success



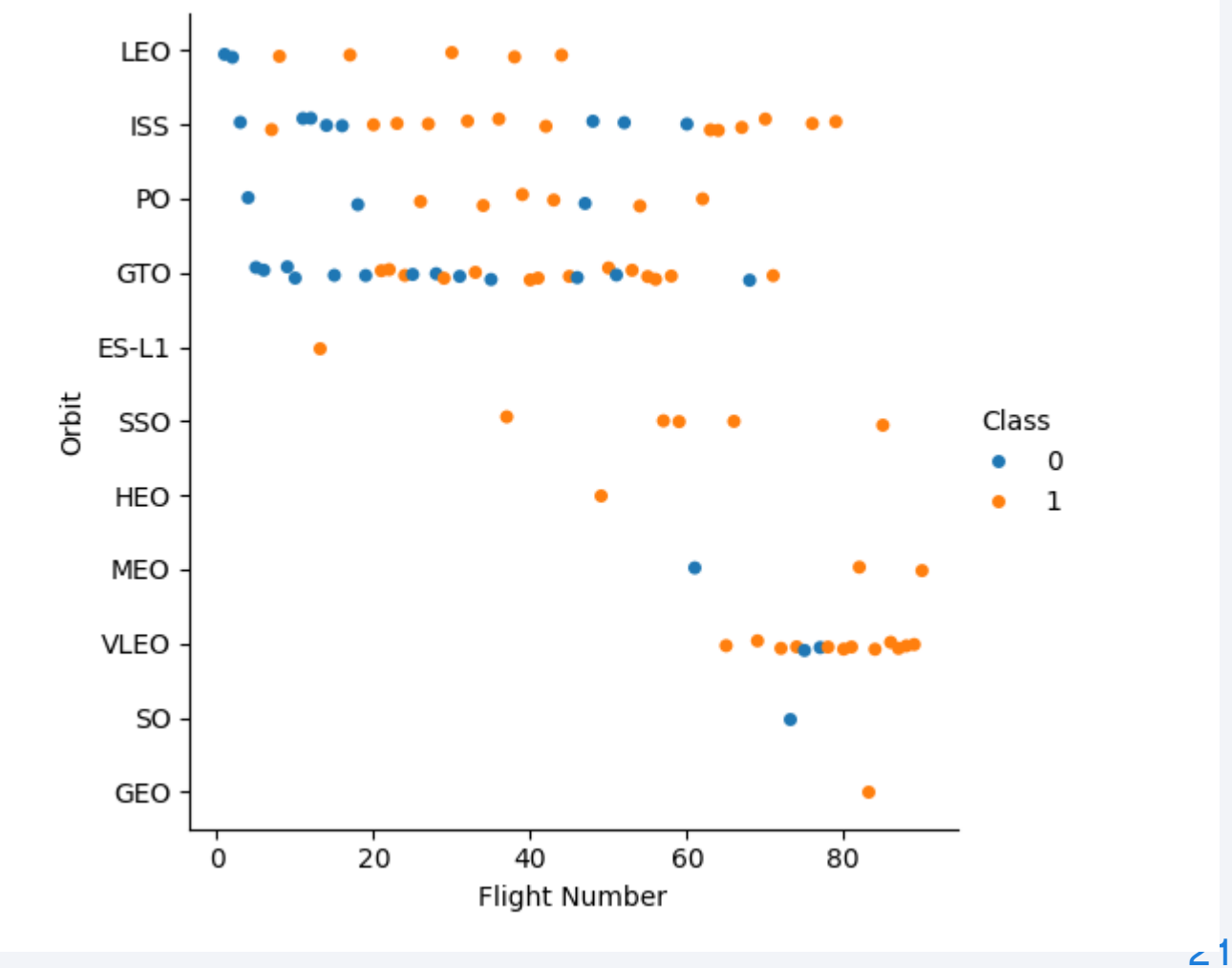
Success Rate vs. Orbit Type

- The bar chart displays average success rate by orbit reached
- ES-L1, GEO, HEO, and SSO orbits have about 100% success rates on launches
- All orbits other than SO have a success rate of at least 50%



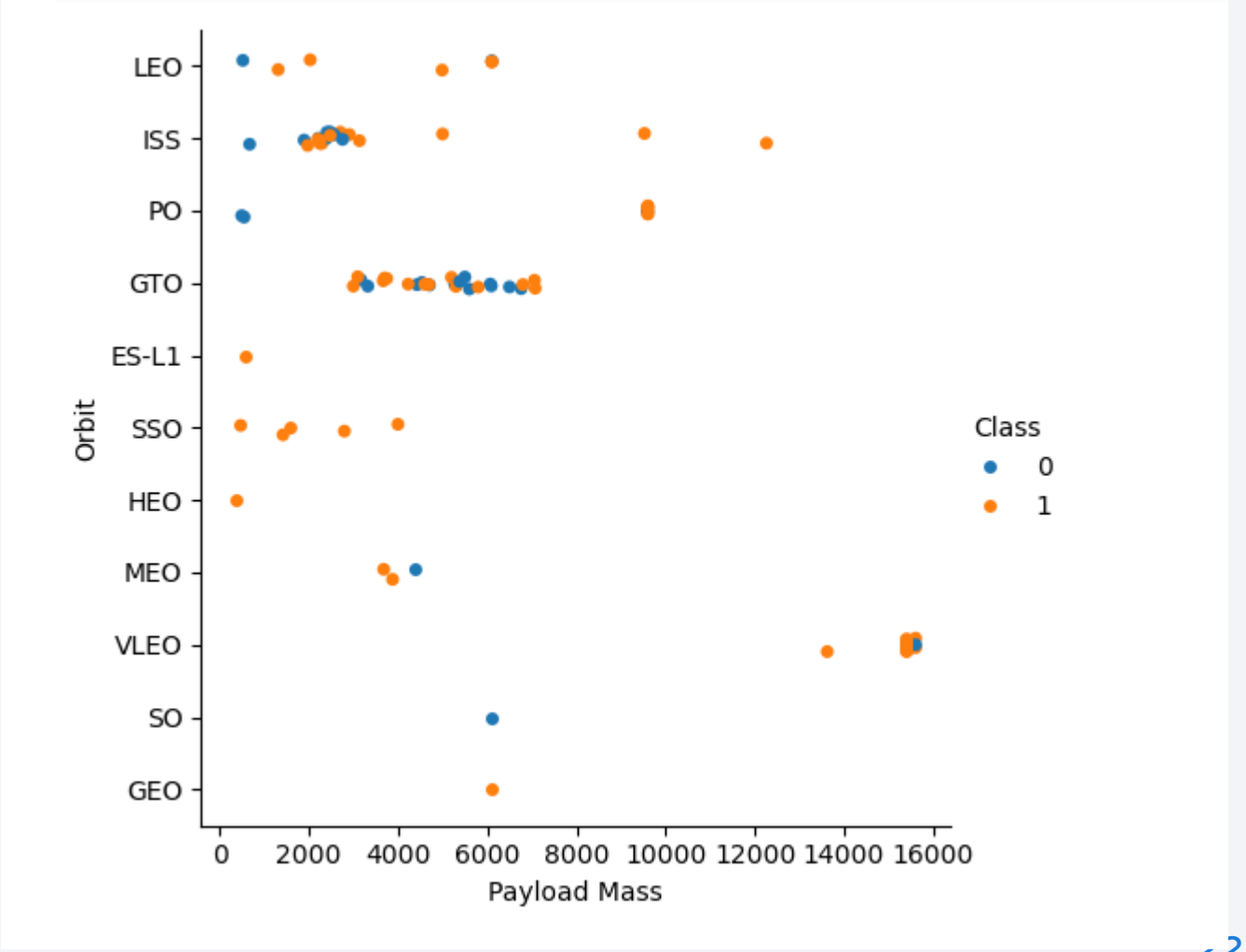
Flight Number vs. Orbit Type

- Flight number and successful launches compared with the orbit that the reached
- The GTO orbit appears to have one of the lowest success rates



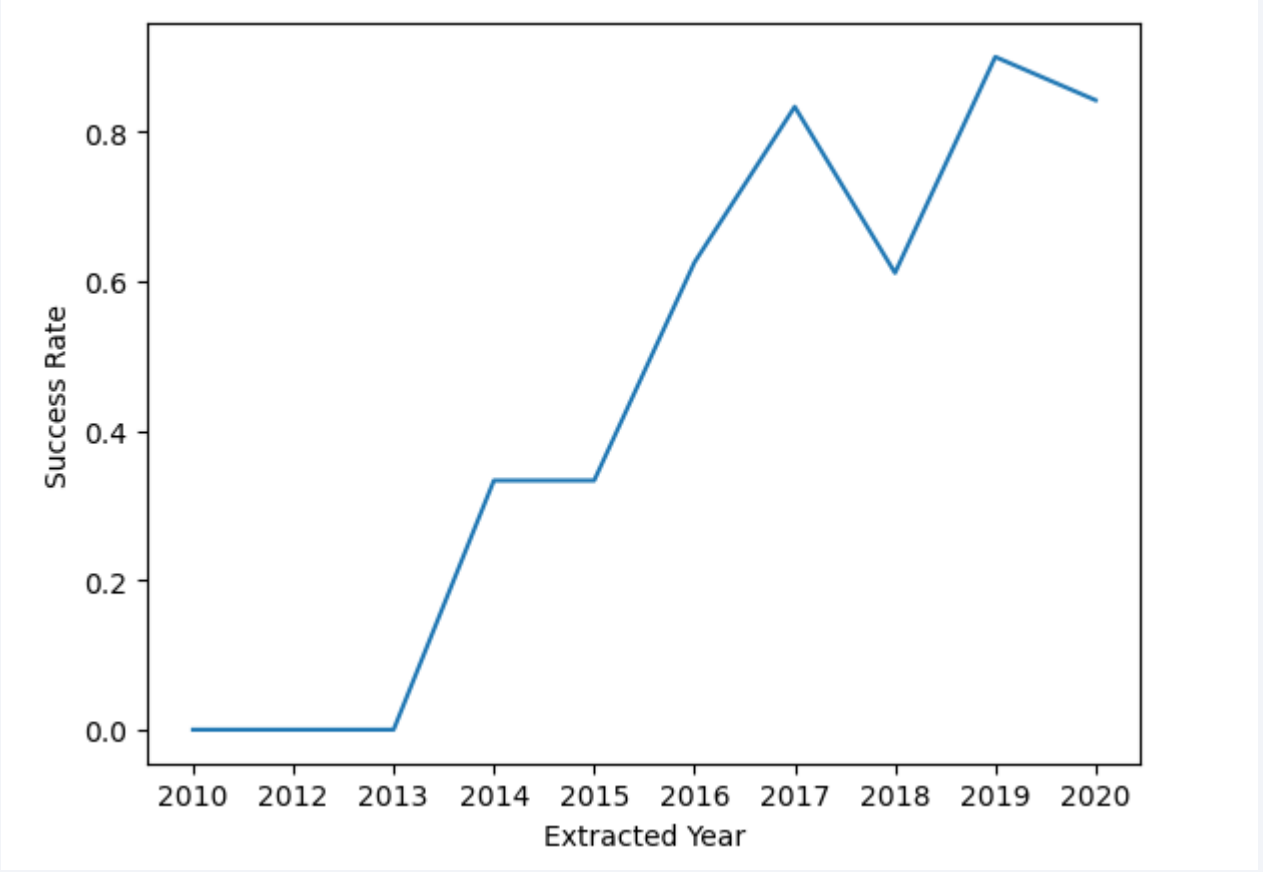
Payload vs. Orbit Type

- Payload mass compared with orbit reached in successful and failed launches
- Payload mass seems to be correlated to the orbit reached meaning certain payload masses can be carried to certain orbits



Launch Success Yearly Trend

- Success rate for launches can be visualized over a 10 year span between 2010 and 2020
- The success rate sharply increases after 2013, while slightly dropping in 2018 before continuing to climb



All Launch Site Names

- The SpaceX launch data consists of four launch sites: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, and CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Five sites were extracted beginning with ‘CCA’ which filtered down results to the launch site CCAFS LC-40

Total Payload Mass

- The total payload mass carried by boosters launched by Nasa was 99,980 kilograms, and found by summing the payload mass column

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 is about 2534.67 kilograms
- The value was extracted by finding the average of the payload mass column

First Successful Ground Landing Date

- The first successful landing outcome in ground pad was achieved on December 22, 2015
- This was found by extracted the minimum date from the date column where the landing outcome was successful

Successful Drone Ship Landing with Payload between 4000 and 6000

- The following boosters had a successful drone ship landing and payload mass between 4000 and 6000:
  - F9 FT B1022
  - F9 FT B1026
  - F9 FT B1021.2
  - F9 FT B1031.2

  - This result was found by filtering the landing outcome column for values similar to drone and success to find only successful landings on drone ships
  - Results were filtered down again to only include values between 4000-6000

Total Number of Successful and Failure Mission Outcomes

- There were 61 successful missions outcomes and 10 failed

Boosters Carried Maximum Payload

- The following boosters were extracted by filtering for the maximum payload mass:

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- The following two records were retrieved for failed landing outcomes on a drone ship in 2015:
  - In January 2015, booster version F9 V1.1 B1012 failed from the CCAFS LC-40 launch site
  - In April 2015, booster version F9 V1.1 B1015 failed from the CCAFS LC-40 launch site

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- The following results are the occurrences of landing outcome by location:
  - 10 landings were not attempted
  - 5 successful landings occurred on drone ships while 5 failed
  - 3 successful landings occurred on ground pads
  - 3 controlled landings occurred in the ocean while 2 uncontrolled occurred
  - 2 landings by parachute failed
  - One landing was precluded on a drone ship


Section 3
Launch Sites
Proximities Analysis

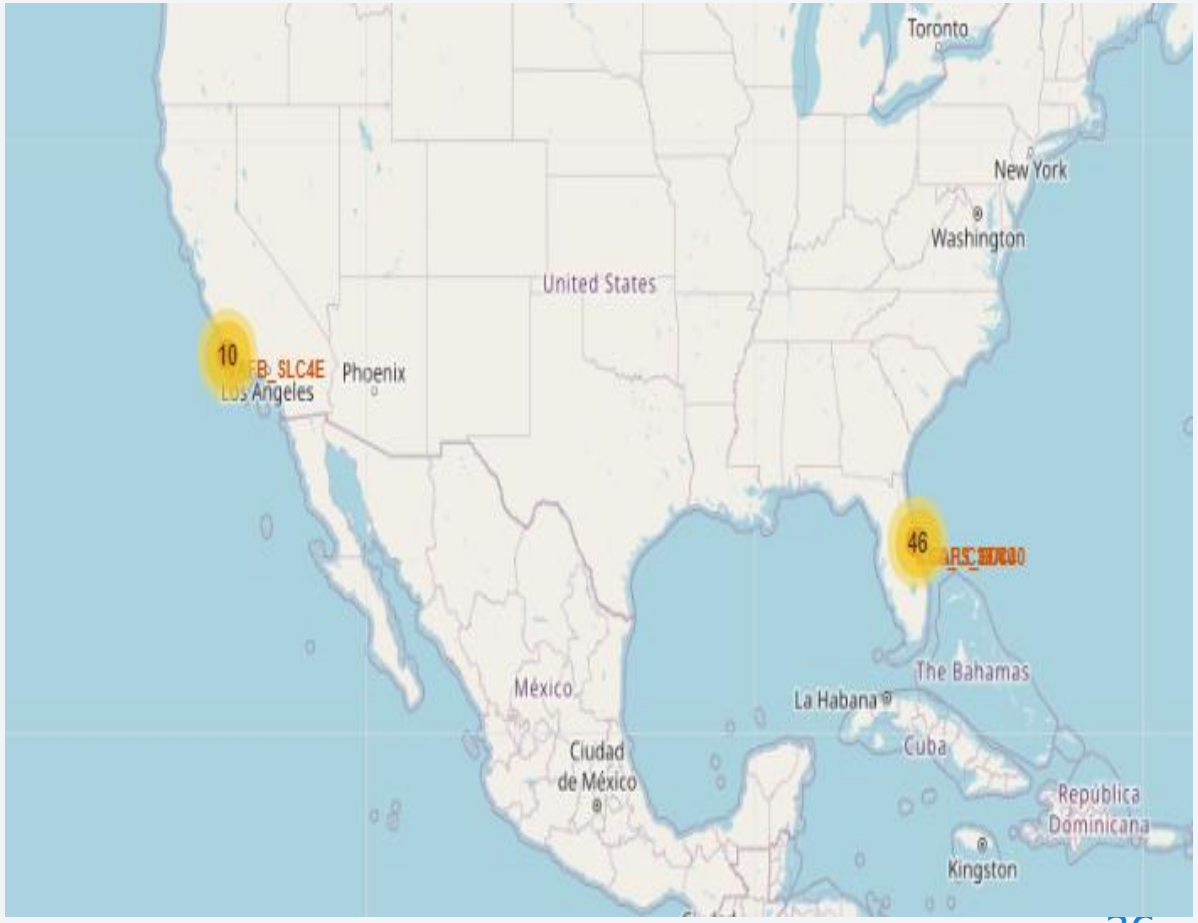
Launch Site Locations

- The map displays the four launch sites for the F9 booster version
- Three launch sites are located in Florida while one is in California
- All launch sites are near a coast line



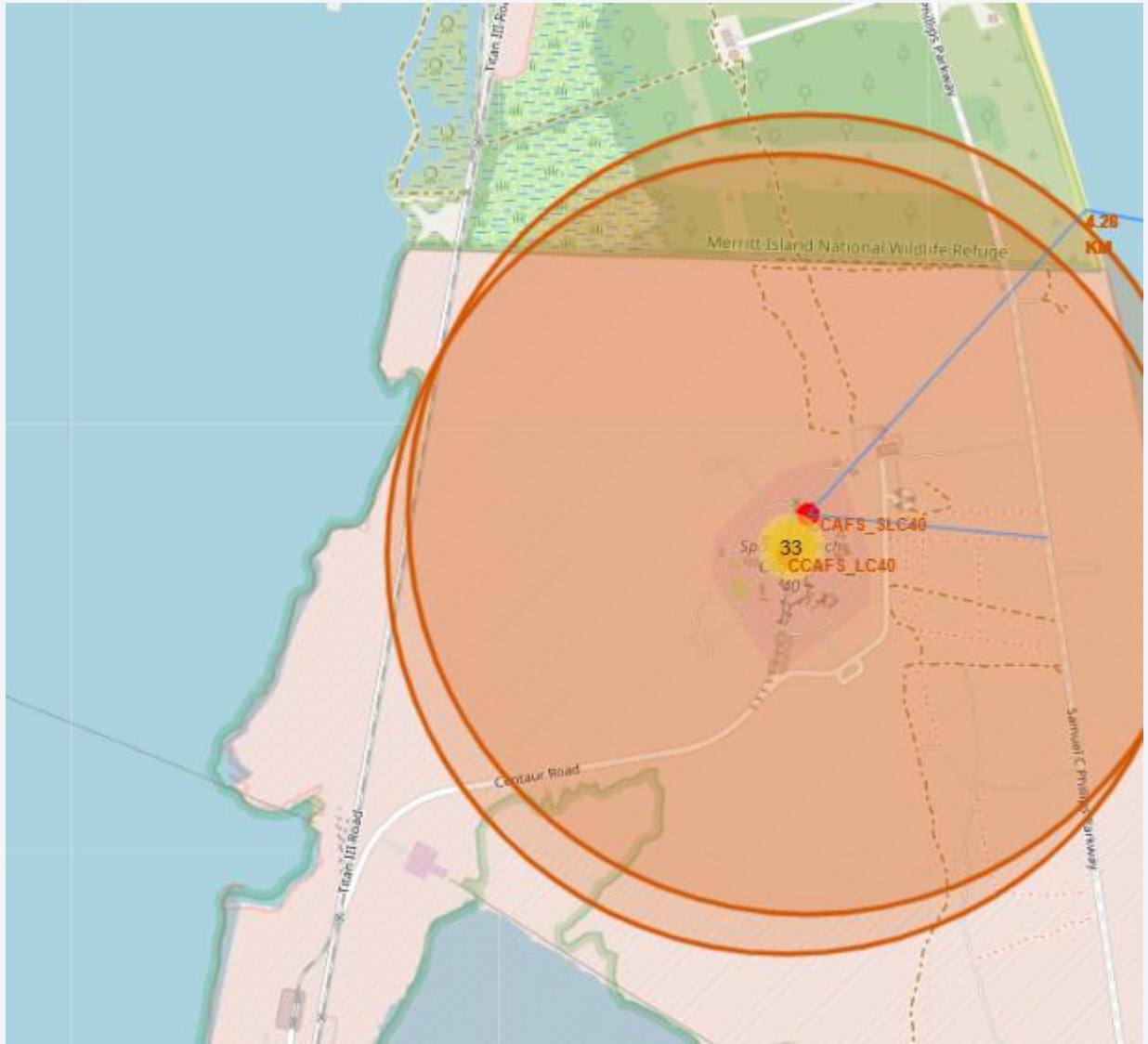
Launch Site Missions

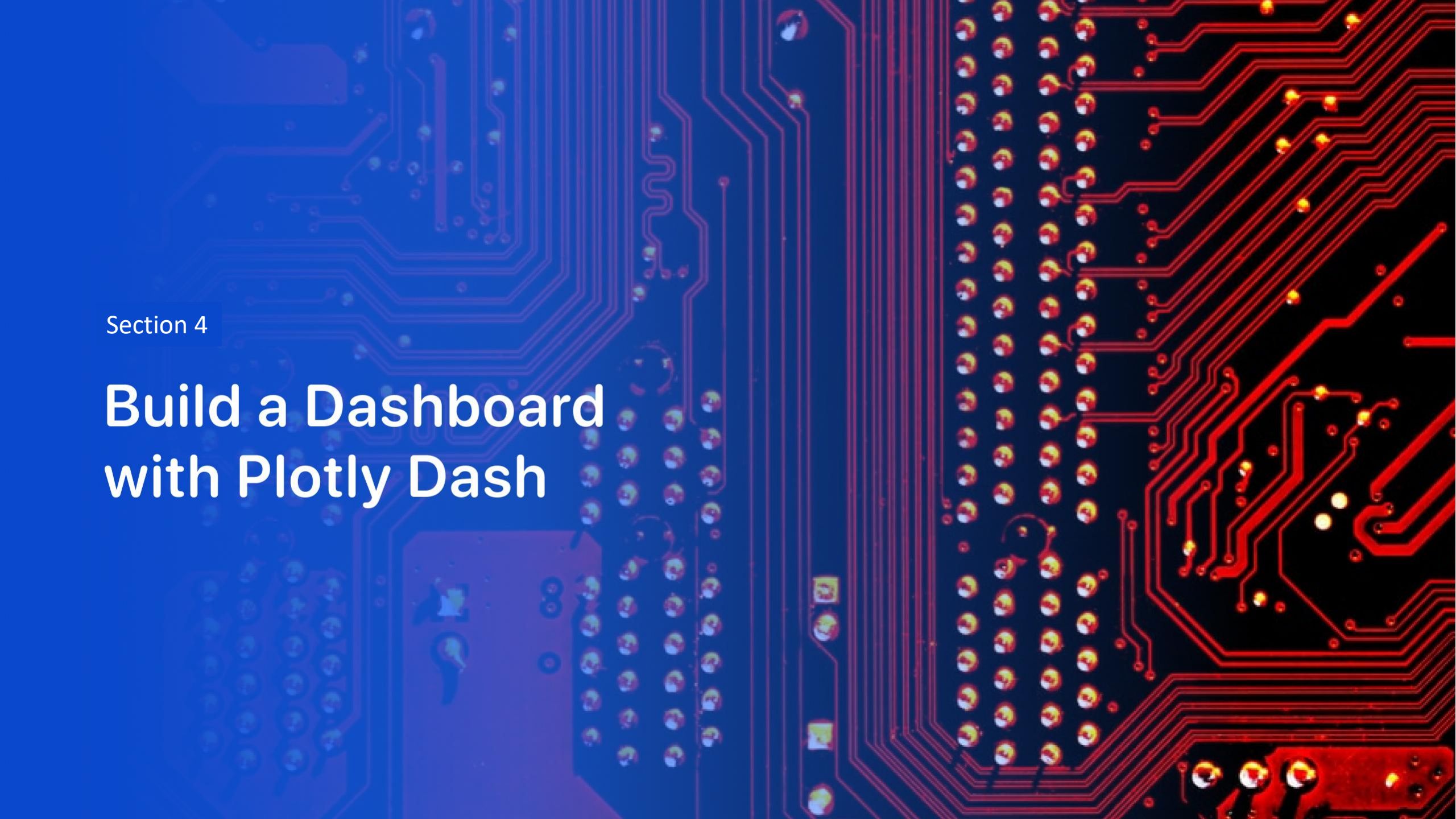
- The map was altered to mark launch locations while also showing green for successful launches and red for failed outcomes
- Additionally, markers show the total number of launches



Distance to Coastline

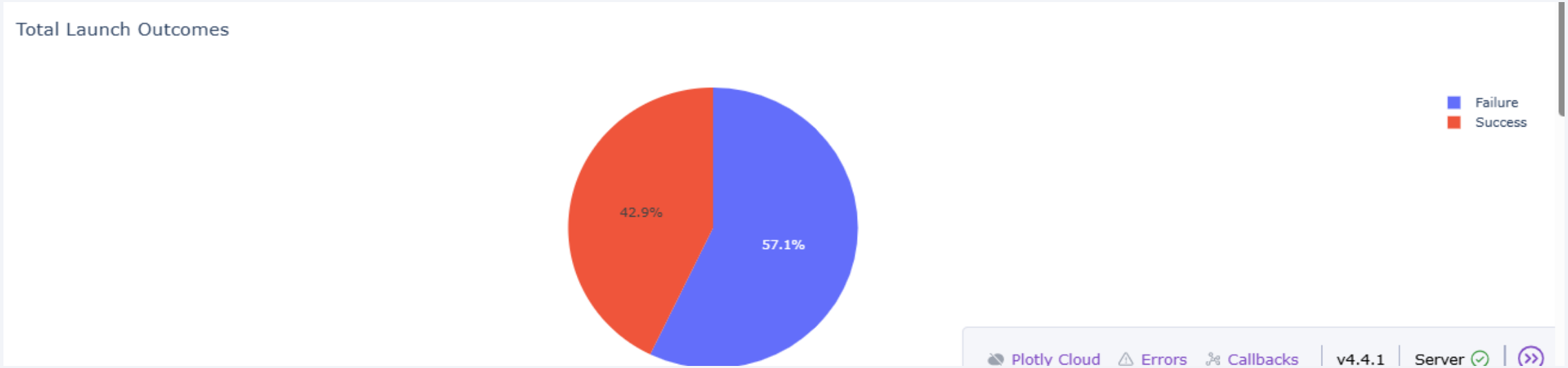
- The screenshot of the Folium map displays the distance of site CAFS SLC-40 to the coastline
- Upon increasing visibility, the site also shows a launch number of 7




Section 4
Build a Dashboard with Plotly Dash

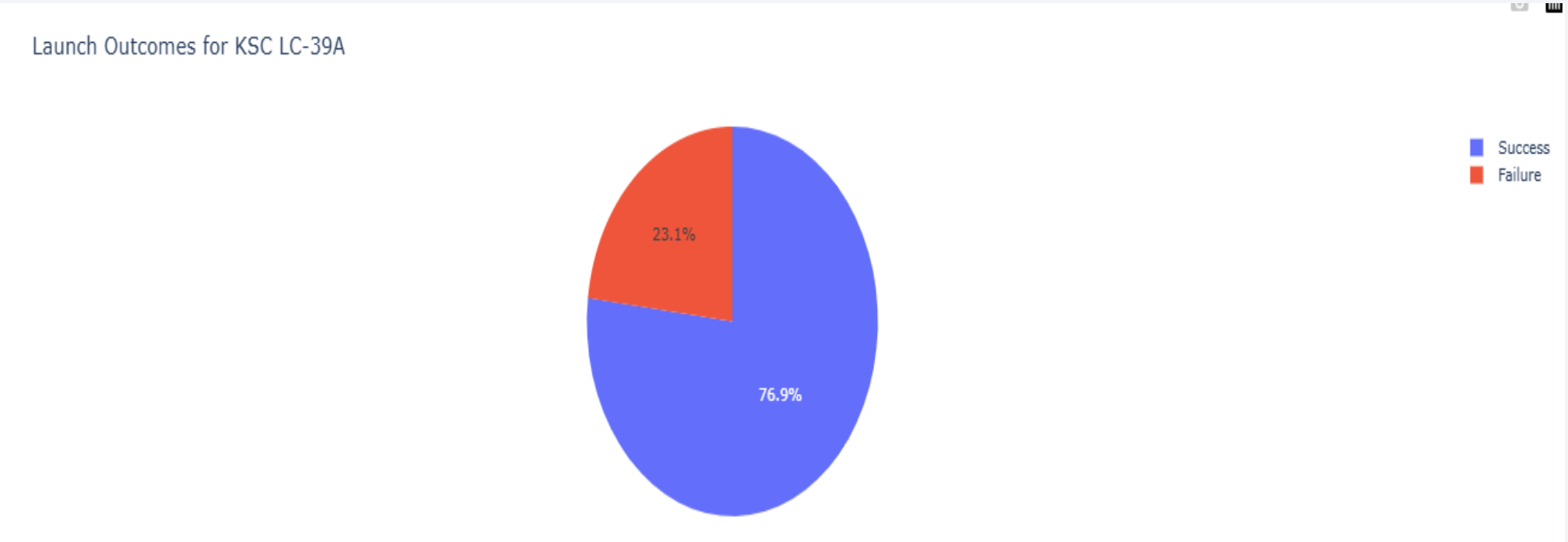
Landing Outcomes

- The pie chart represents the total percentage of successful and failed landings
- The success rate of the F9 booster is 57.1%



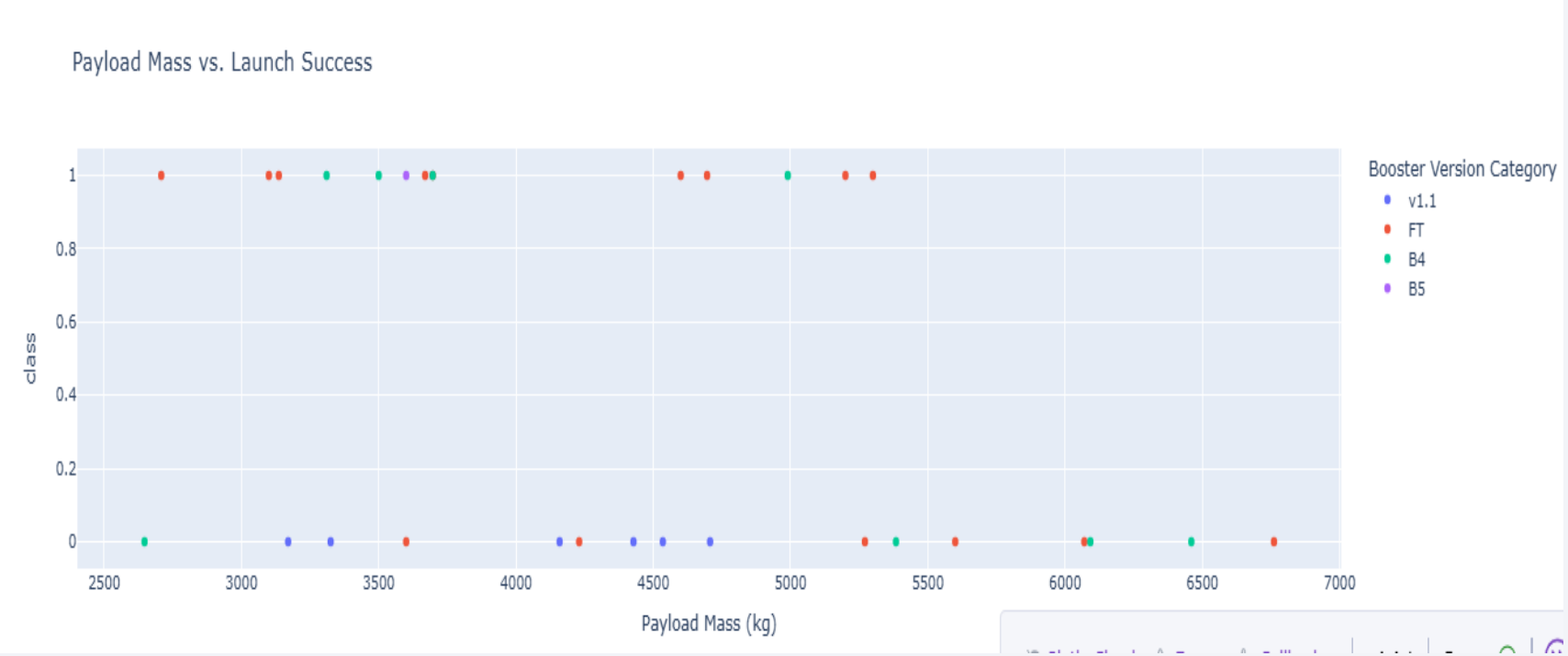
Best Performing Launch Site

- Launch site KSC LC-39A had the highest successful launch rate with 76.90%



Booster Performance by Payload Mass

- The following screenshot of the dashboard payload mass range slider indicates the booster versions for all launch sites which carried a payload mass of 2500-7500 kilograms and their associated success rates
- The FT booster seems to be the most successful at this mass



Section 5

Predictive Analysis (Classification)



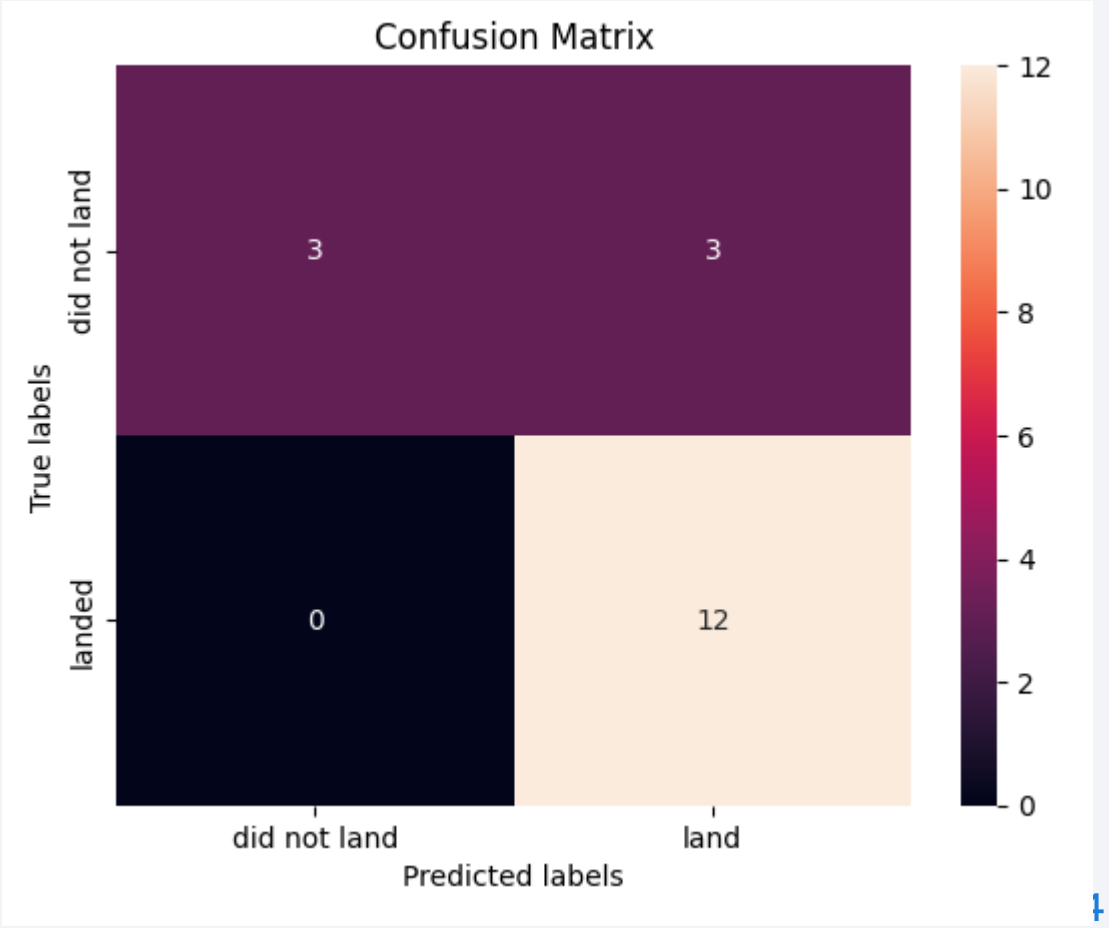
Classification Accuracy

- On testing data the logistic regression, K-nearest neighbors, and supervised models perform at an accuracy rate of 83.33%
- On testing data the decision tree performed the worst ith an accuracy rate of 72.22%

- The following are the accuracy measures on model s for training data
  - K-nearest neighbors: 82.82%
  - Decision tree: 87.50%
  - Supervised model: 84.82%
  - Logistic regression: 84.64%

Confusion Matrix

- The confusion matrix is the same for the three mentioned models
- 12 successful landings were accurately predicted
- 3 unsuccessful landings were accurately predicted
- 3 unsuccessful landings were inaccurately predicted as being successful



Conclusions

- Exploratory analysis found possible relationships between payload mass and successful launches into certain orbits
- Exploratory analysis also found that launches began to have success in 2013 and increased through to 2020
- The first successful landing occurred in 2015
- The interactive map displays all launch sites are near coastlines
- Data from all launch sites averaged to a 57.1% success rate on landings
- The supervised model seems to be the most successful model with 83.33% accuracy on test data and 84.82% accuracy on training data

Thank you!

